



INTRODUCTION

MACHINE LEARNING CAPSTONE

- **Objective:** Develop a personalized course recommender system for an online learning platform.
- **Business Problem:**
 - Large number of available courses.
 - Learners struggle to discover relevant content.
 - Personalized recommendations improve learning experience.
- **Solution:**
 - Compare multiple recommendation approaches
 - Content-Based Filtering
 - Collaborative Filtering
 - Neural Network-based Recommendation

COURSE COUNTS PER GENRE

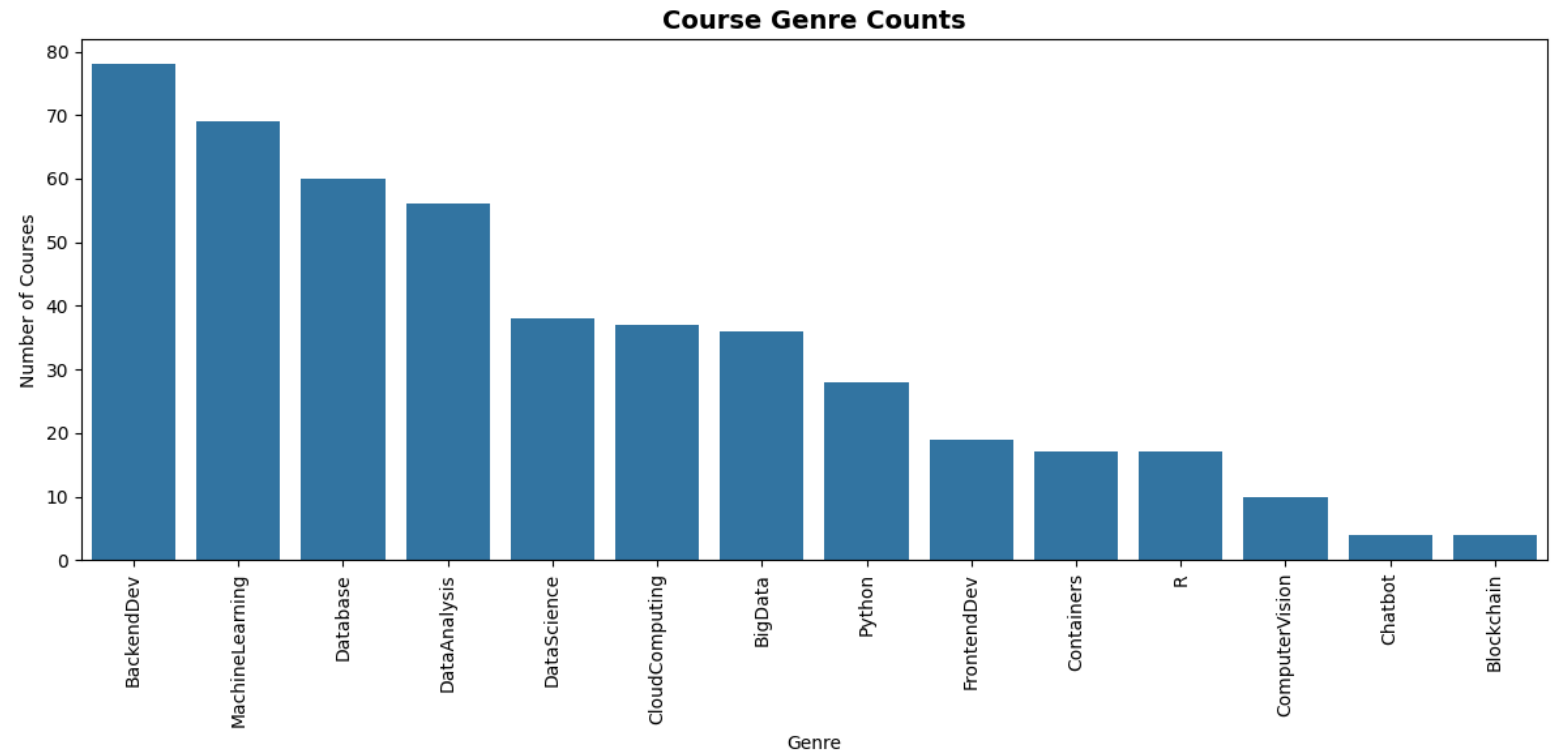
Objective: Understand how courses are distributed across different genres.

Method: Course genre columns were summed to count how many courses belong to each category.

Key Findings:

- Backend Development has the highest number of courses.
- Machine Learning, Database, Data Analysis, and Data Science are also major categories.
- The course catalog is strongly focused on technical and AI-related topics.

Conclusion: Genre distribution helps identify popular learning areas and supports better course recommendation.



USER ENROLMENT DISTRIBUTION / EDA SUMMARY

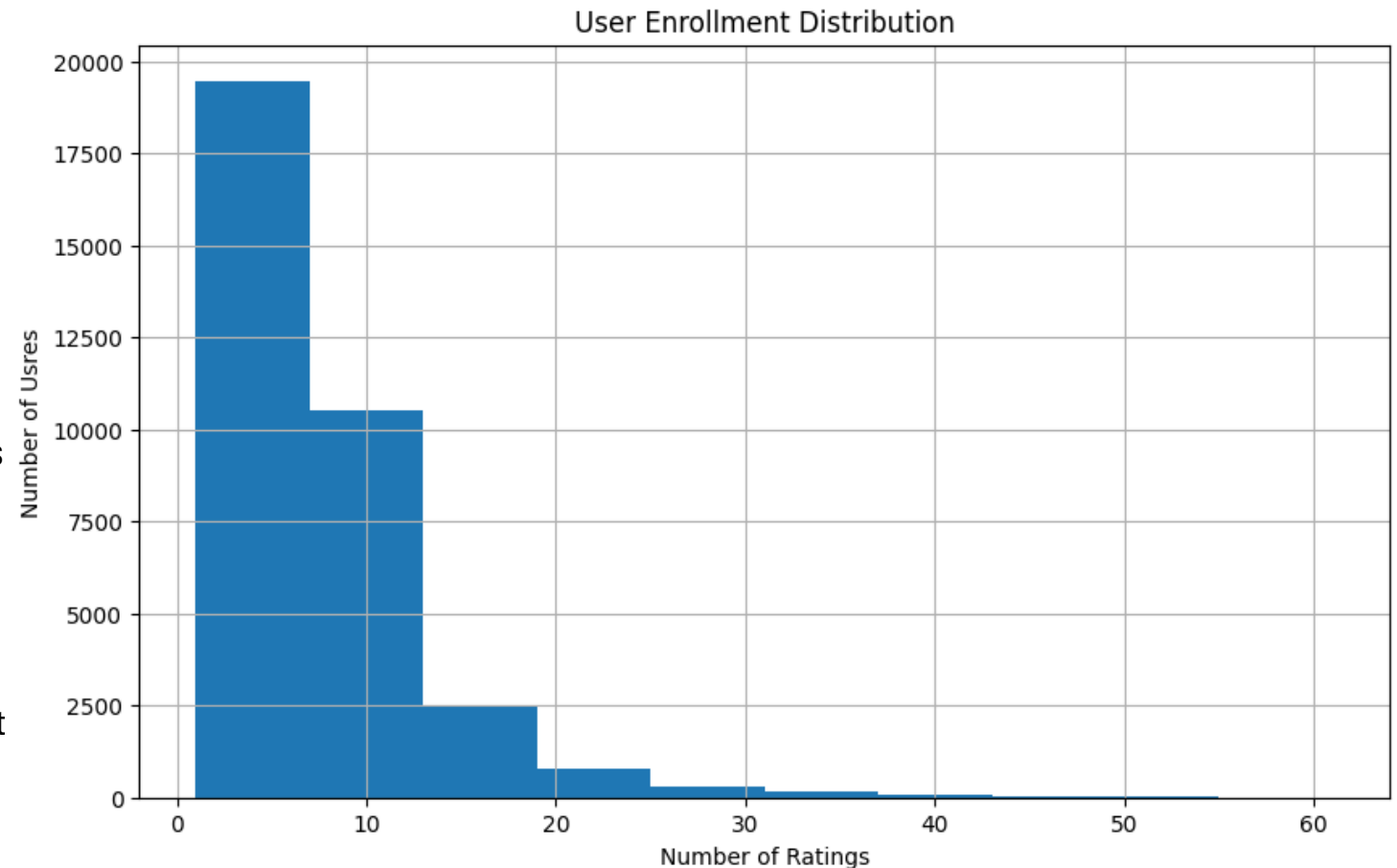
Objective: Understand learner enrolment behaviour and course activity patterns.

Method: Grouped enrolment/rating data by user to count how many courses each learner enrolled in.

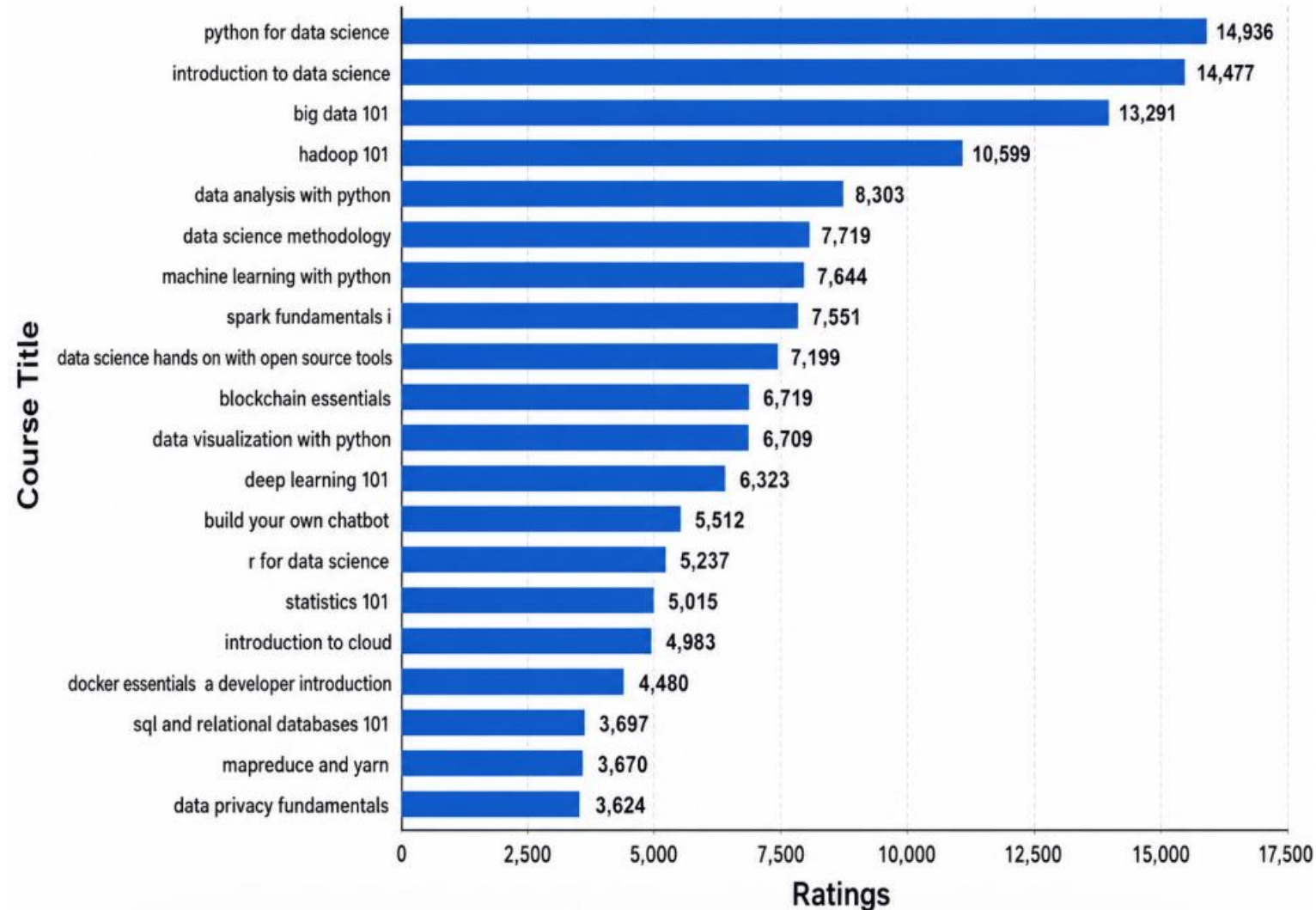
Key Findings:

- Most learners enrolled in only a few courses.
- A smaller group of learners enrolled in many courses
- Course activity is not evenly distributed across all users.

Conclusion: This pattern shows the need for personalized recommendations to help learners discover more relevant courses and improve engagement.



Top 20 Most Popular Courses by Enrollment



TOP 20 MOST POPULAR COURSES BY ENROLMENT

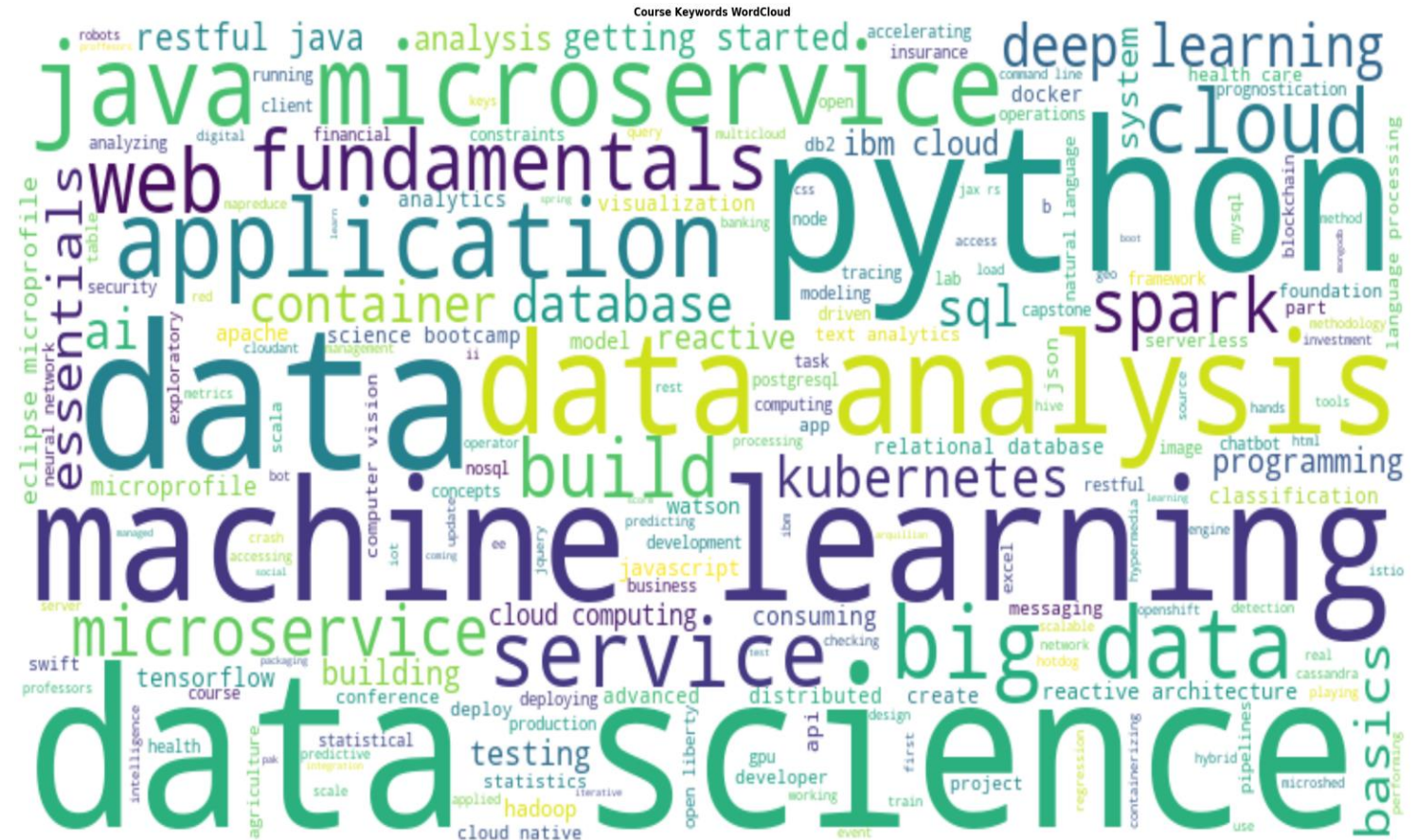
WORD CLOUD VISUALIZATION

Purpose: Identify the most frequent keywords in course titles.

Key Terms: Python, Data Science, Machine Learning, Data Analysis, Cloud, Java, Spark.

Insight: Larger words appear more often in the course catalog.

Conclusion: The catalog is strongly focused on Python, Data Science, Machine Learning, and Cloud-related topics.



USER PROFILE RECOMMENDER: FLOWCHART & EVALUATION

Objective: Recommend courses based on learner history and course genre preferences.

Methodology:

- Build user profile from completed courses.
- Match user profile with unseen course genre vectors.
- Calculate recommendation scores.
- Rank and recommend top courses.

Flowchart: Completed Courses → Course Genre Matrix → User Profile Vector → Recommendation Scores → Top-NCourses

Evaluation:

- Personalized and easy to interpret.
- Fast recommendation generation.
- Depends on learner history and genre quality.
- Cold start issue for new users.

Key Finding: User interests improve personalization by recommending courses aligned with learner preferences.

COURSE SIMILARITY WORKFLOW & EVALUATION

This recommender suggests courses that are similar to the courses a learner has already viewed or completed. Course descriptions are converted into text features, and cosine similarity is used to measure how closely two courses are related. The system then selects the top similar courses and recommends them to the learner.

Workflow: Course Descriptions → Text Features → Cosine Similarity Matrix → User's Last Interacted Courses → Top-N Similar Course Recommendations

Evaluation: The course similarity approach gives highly relevant recommendations when courses share similar technical content and keywords. It performs well for recommending closely related topics, but it may also create a filter bubble by repeatedly suggesting similar subjects and missing new or diverse learning areas.

COURSE SIMILARITY RECOMMENDER: EVALUATION RESULTS

Objective: Evaluate how well the course similarity approach recommends related courses.

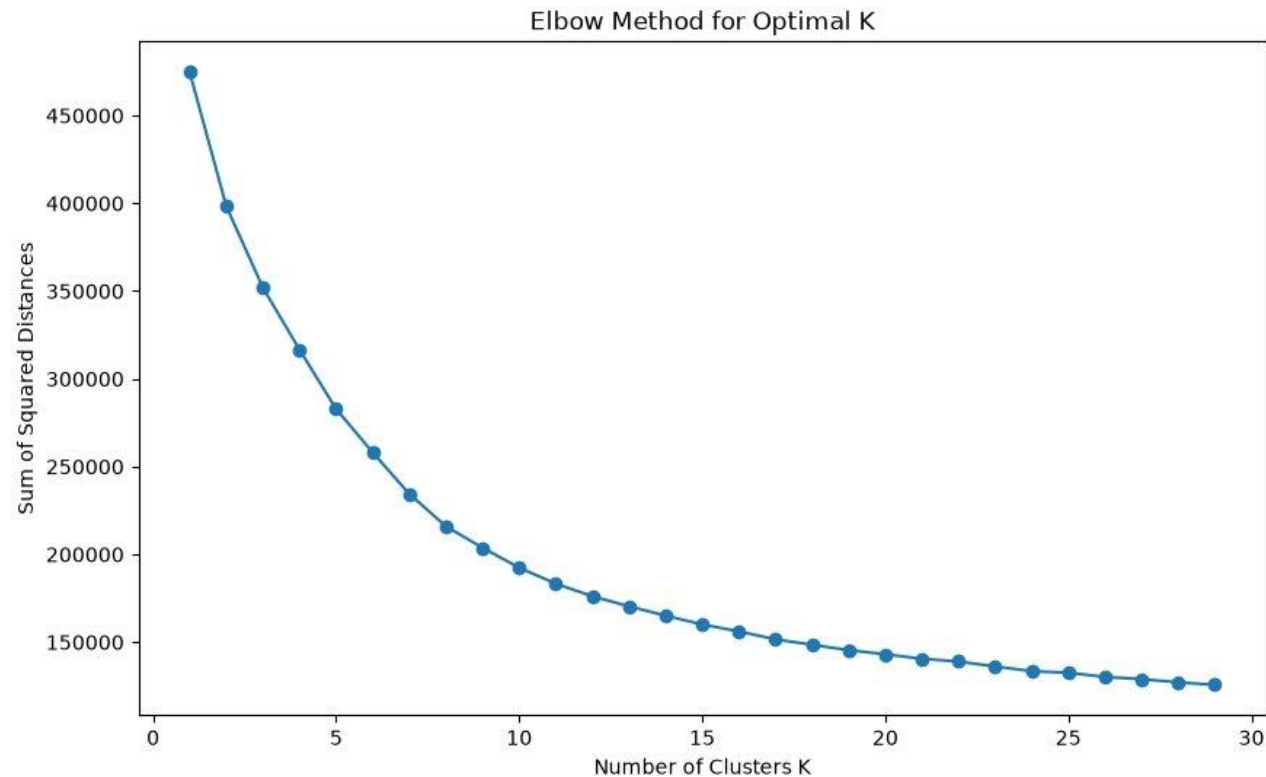
Evaluation Summary:

Evaluation Area	Result
Feature Type	Bag of Words course text features
Similarity Method	Cosine Similarity
Input	Users completed or viewed courses
Output	Top-N similar course recommendations
Strength	Recommends closely related technical courses
Limitation	May recommend similar topics repeatedly

Key Finding: The course similarity recommender works well when courses share common technical keywords and descriptions.

Conclusion: This method is fast, simple, and useful for recommending related courses, but it may reduce recommendation diversity.

CLUSTERING RECOMMENDER: FLOWCHART & EVALUATION



- The clustering recommender groups similar learners based on their course interaction patterns. PCA is used to reduce high-dimensional user features, and KMeans is applied to create learner clusters. For each learner, the system recommends popular courses from the same cluster that the learner has not already enrolled in.
- **Workflow:** User-Course Interaction Data → PCA Dimensionality Reduction → KMeans Clustering → Learner Groups → Popular Courses in Cluster → Unseen Course Recommendations
- **Evaluation:** This approach is scalable and useful for learner segmentation. It improves personalization by recommending courses based on the behavior of similar users in the same cluster. However, the quality depends on choosing the right number of clusters and having enough user interaction data.
- **Key Finding:** Clustering helps identify groups of similar learners and recommend relevant courses based on shared learning patterns.

CLUSTERING EVALUATION RESULTS

Objective: Evaluate how well clustering groups similar learners and supports course recommendation.

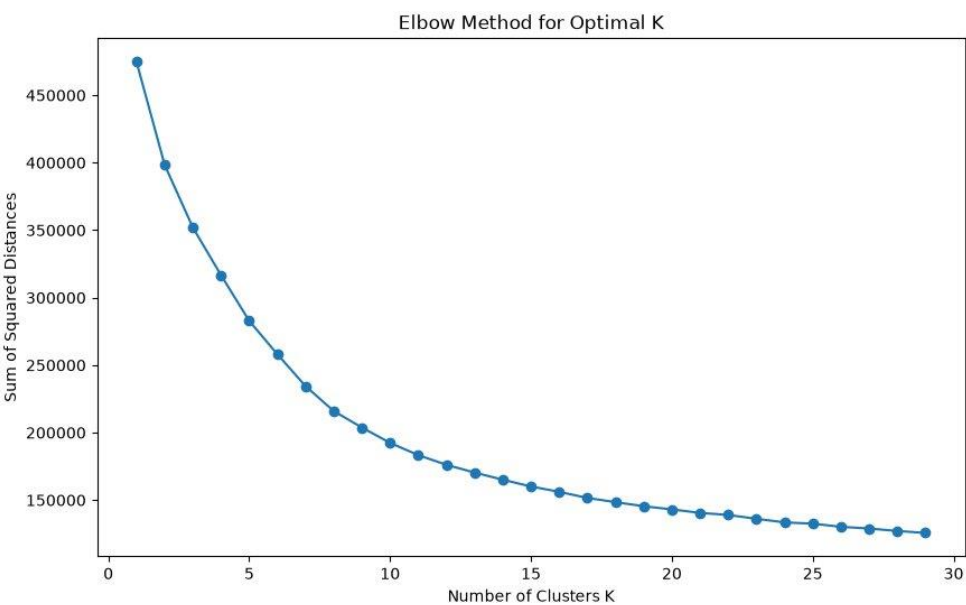
Method Used: PCA + KMeans Clustering

Evaluation Summary:

Evaluation Area	Result
Input Data	User-course interaction features
Dimensionality Reduction	PCA
Clustering Algorithm	KMeans
Evaluation Method	Elbow Method
Recommendation Logic	Recommend popular unseen courses from the same cluster
Strength	Groups similar learners effectively
Limitation	Performance depends on selecting the right number of clusters

Key Finding: The clustering approach helps segment learners into similar groups. After users are grouped, popular courses within the same cluster can be recommended to learners who have not yet enrolled in those courses.

Conclusion: Clustering is useful for learner segmentation and scalable recommendations, especially when users share similar course interaction patterns.



COLLABORATIVE FILTERING OVERVIEW

Collaborative filtering is a technique used by recommendation systems to predict a user's interests by finding patterns in the behavior and preferences of similar users. It works on the core principle that people who agreed on preferences in the past will agree in the future

- KNN
 - Find nearest users.
 - Predict ratings.
- NMF
 - Matrix factorization.
 - Learn latent features.
- Key Findings
 - KNN is simple and interpretable.
 - NMF handles sparse matrices effectively.
- Evaluation: RMSE

KNN-BASED RECOMMENDER SYSTEM FLOWCHART

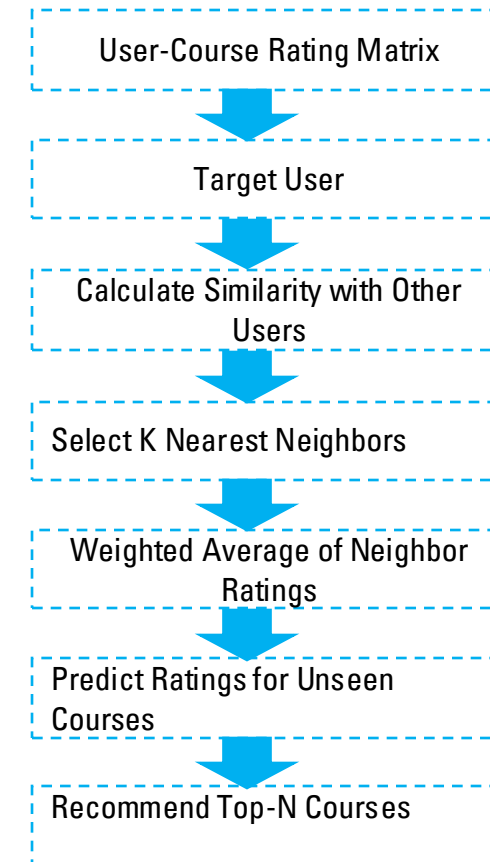
Objective: Predict ratings for unseen courses by finding learners with similar course preferences.

Evaluation:

- Metric used: RMSE
- Lower RMSE indicates better rating prediction performance.

Key Finding: KNN is simple and interpretable because recommendations are based on similar learners' behavior.

Workflow:



NMF-BASED COLLABORATIVE FILTERING WORKFLOW

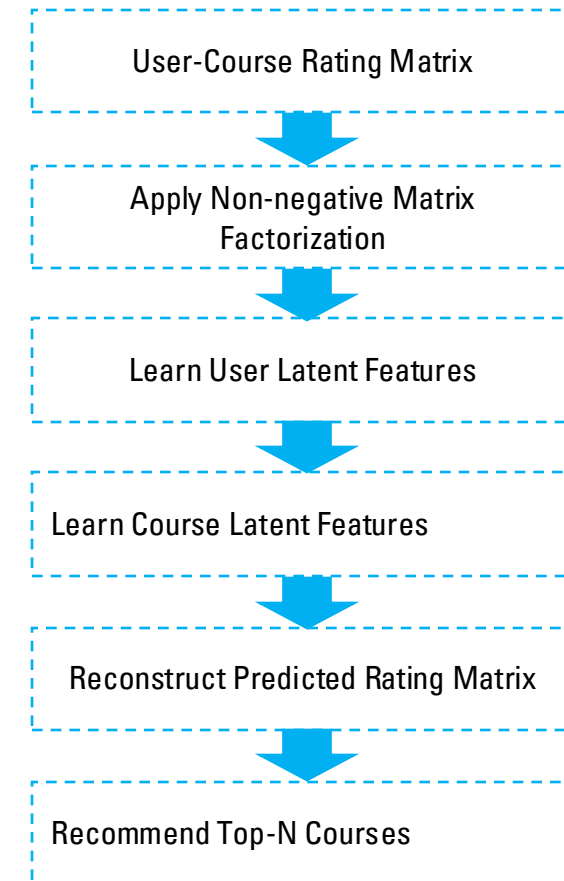
Objective: Predict learner-course ratings by discovering hidden user and course features.

Evaluation:

- Metric used: RMSE
- Lower RMSE means better rating prediction.

Key Finding: NMF handles sparse user-course interaction data effectively and learns hidden patterns between learners and courses.

Workflow:



COLLABORATIVE FILTERING MODEL COMPARISON

Objective: Compare collaborative filtering models used for course recommendation.

Model	Method	Evaluation Metric	Strength
KNN	Finds similar users	RMSE	Simple and easy to interpret
NMF	Matrix factorization	RMSE	Handles sparse data effectively
Neural Network	Embedding-based learning	Loss / RMSE	Learns hidden user-course relationships

Key Finding: KNN is easier to understand, while NMF works better with sparse user-course rating data. Neural networks provide deeper personalization by learning hidden patterns from user and course embeddings.

Conclusion: Collaborative filtering models improve recommendations by using learner interaction history instead of only course content.

NEURAL NETWORK RECOMMENDATION

- Method
 - User embeddings.
 - Course embeddings.
 - Dense neural network.
- Key Findings
 - Learns hidden relationships.
 - Captures latent features.
 - Improves recommendation quality.

A Neural Network Recommendation is an advanced AI system that predicts user preferences and suggests relevant items (like movies, products, or videos) by analyzing complex, non-linear patterns in data using layers of artificial neurons.

NEURAL NETWORK EMBEDDING WORKFLOW

Workflow: User ID → User Embedding Layer Course ID → Course Embedding Layer

User Embedding + Course Embedding → Interaction / Concatenation Layer → Dense Neural
NetworkLayers → Predicted Rating / Preference Score → Course Recommendation

Evaluation:

- The model is evaluated using prediction loss and RMSE.
- Lower RMSE indicates better rating prediction performance.

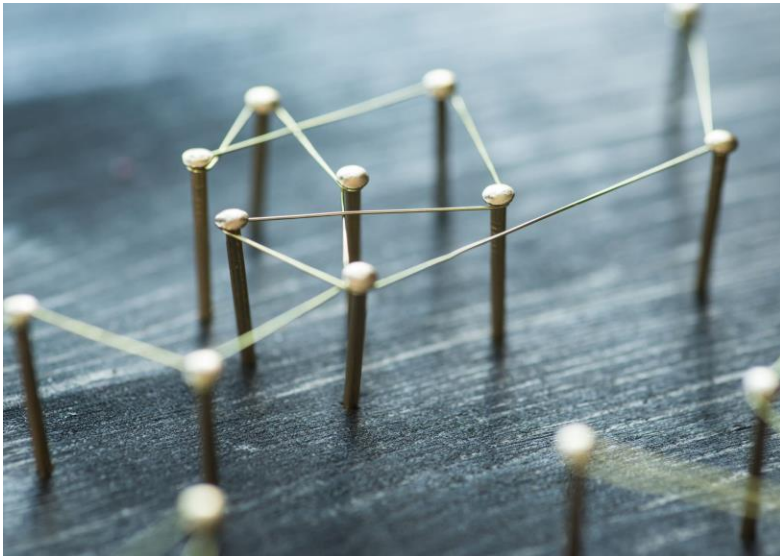
Key Point: The embedding workflow explains how sparse user and course IDs are converted into dense vectors and used by the neural network to generate personalized recommendations.

RATING PREDICTION

- Regression
 - Predict rating scores.
 - Linear Regression.
- Classification
 - Predict rating modes.
 - Logistic Regression.
 - Random Forest.
- Evaluation
 - RMSE
 - Accuracy
 - Precision
 - Recall
 - F1 Score
- Key Findings: Embedding features improve prediction quality.

Rating prediction is a machine learning technique used to estimate the numerical score (e.g., 1 to 5 stars) a user would give an unseen item or service. It is the foundation of modern recommendation systems in e-commerce and streaming, personalizing user experiences by anticipating preferences.

MODEL COMPARISON



- Model Strength
- User Profile Personalized
- Course Similarity Fast
- Clustering Learner segmentation
- KNN Easy to interpret
- NMF Sparse matrix handling
- Neural Network Hidden relationships
- Evaluation Metrics
 - RMSE
 - Accuracy
 - Recommendation quality
- Key Finding: Neural networks create deeper, more meaningful representations of courses/users than simpler methods.

PROJECT WORKFLOW



- Course Data
- EDA
- Bag of Words
- Course Similarity
- User Profiles
- Content-Based Recommendation
- Clustering
- KNN
- NMF
- Neural Networks
- Regression
- Classification
- Final Recommendation System

CONCLUSION

- Built
 - Content-Based Recommender.
 - User Profile Recommender.
 - Clustering Recommender.
 - KNN Collaborative Filtering.
 - NMF Collaborative Filtering.
 - Neural Network Recommender.
- Key Findings
 - AI-related courses dominate learner interests.
 - Personalized recommendations improve discovery.
 - Neural networks capture complex user-course relationships.
- Future Work
 - Hybrid recommender systems, Streamlit deployment.
 - Cloud deployment, Real-time recommendations.

This project successfully built and evaluated a diverse suite of recommendation engines, from foundational content-based models to advanced neural networks.

The findings highlight a strong learner preference for AI courses and prove that personalized models significantly improve content discovery, with neural networks proving best at capturing complex user behavior.

To build on these achievements, future work will focus on combining these models into a hybrid system and deploying the architecture to the cloud via Streamlit to deliver scalable, real-time recommendations.